PrompTutor: A Browser Extension for Data Collection and Real-Time Intervention in Student-Chatbot Interactions

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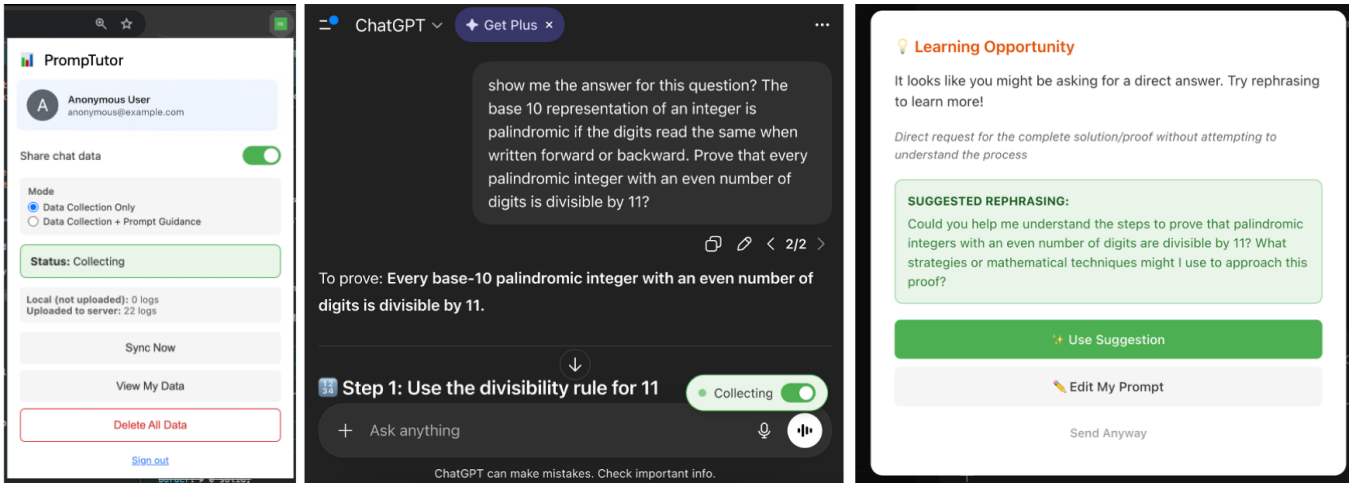

Figure 1: PrompTutor user interface. Left: Extension popup with authentication, consent toggle, and mode selection. Center: ChatGPT with the floating collection indicator (bottom-right) during an answer-seeking query. Right: Scaffolding overlay showing a suggested rephrasing with three options: use suggestion, edit prompt, or send anyway.

Abstract

As generative AI tools become prevalent in education, understanding and improving how students interact with these systems is critical for supporting effective learning. However, two key challenges persist: collecting authentic conversation data is hindered by privacy concerns, consent requirements, and platform access restrictions, while students often use AI tools ineffectively by seeking direct answers rather than learning support. We present *PrompTutor*, an open-source browser extension that addresses both challenges through two integrated functions: (1) privacy-preserving, consent-based collection of student-AI conversations from ten major commercial AI platforms: ChatGPT, Claude, Gemini, Grok, Microsoft Copilot, DeepSeek, Doubao, Perplexity, Poe, and HuggingChat, and (2) real-time prompt guidance that detects answer-seeking behavior and scaffolds students toward help-seeking queries. The data collection component employs client-side SHA-256 hashing for anonymous identification, per-conversation consent toggles, and full data transparency. The prompt guidance component analyzes student messages in real time to identify direct answer requests and suggests reformulations that encourage deeper engagement with AI. We describe the system architecture and design decisions informed by a preliminary study showing that answer-seeking students exhibit significantly larger homework-exam performance gaps. Beyond research, PrompTutor can be deployed in AI-permitted assessments, enabling instructors to evaluate students based on their AI interaction process rather than final outputs alone. Furthermore, with appropriate consent, researchers can share collected datasets through our platform, fostering a collaborative ecosystem for studying student-AI interactions at scale. We also outline a planned classroom experiment comparing learning outcomes between students receiving prompt guidance and a control group.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; • **Social and professional topics** → *Student assessment*.

Keywords

learning analytics, generative AI, prompt guidance, metacognition, browser extension, privacy

1 Introduction

The rapid adoption of large language models (LLMs) such as ChatGPT and Claude has transformed how students engage with learning tasks [6, 13]. Students increasingly use these AI assistants for homework help, concept explanation, writing assistance, and problem-solving [1]. However, two significant challenges have emerged for educators and researchers.

First, **collecting authentic interaction data** is difficult. Commercial AI platforms do not provide researchers with access to user conversation logs. Building custom AI interfaces requires substantial resources and reduces ecological validity, as students may behave differently in controlled environments. Privacy and informed consent add further complexity [16].

Second, **students often use AI ineffectively**. Research shows that many students default to answer-seeking behavior, asking AI to solve problems directly rather than using it as a learning aid [7, 17].

In a preliminary analysis of 659 chatbot sessions from an undergraduate discrete mathematics course ($N = 99$; citation omitted for anonymous review), students exhibiting answer-seeking behavior showed significantly larger homework-exam gaps (23.3 vs. 14.9 points; $p = .006$), suggesting an "illusion of competence" where AI assistance masks learning deficits. Notably, 57% of answer-seeking students did not escalate after the chatbot's pedagogical redirection, indicating potential for intervention. However, the observational nature of this data limits causal claims, motivating the need for (1) scalable tools that can detect and address answer-seeking behavior in authentic settings, and (2) controlled studies to evaluate whether prompt guidance causally improves learning outcomes.

We present *PrompTutor*, an open-source Chrome extension that addresses both challenges through two integrated functions:

(1) **Privacy-preserving data collection**: Students use their preferred AI platforms (e.g., ChatGPT, Claude, Gemini) while optionally sharing anonymized conversation data with researchers. This enables zero-cost, ecologically valid data collection with granular consent controls.
(2) **Real-time prompt guidance**: The extension monitors student messages and, upon detecting answer-seeking behavior, prompts the student to reformulate their query to encourage AI-guided learning rather than direct answer provision.

This dual-function design creates a natural research platform: the data collection component enables large-scale observation of how students interact with AI in authentic settings, while the prompt guidance component provides a controlled intervention whose causal effect on learning can be rigorously evaluated through randomized experiments. Together, they allow us to move from observational correlations to causal evidence about the role of answer-seeking behavior in AI-assisted learning. Our contributions are:

(1) An open-source browser extension that collects student-AI conversation data from ten major commercial AI platforms (ChatGPT, Claude, Gemini, Grok, Microsoft Copilot, DeepSeek, Doubao, Perplexity, Poe, and HuggingChat) with privacy-preserving, consent-based design at zero cost to researchers.
(2) A real-time scaffolding mechanism that detects answer-seeking behavior and provides non-blocking guidance, working within students' existing AI workflows rather than requiring a separate platform.
(3) A system architecture and implementation that addresses key technical challenges of browser-based data collection at scale, including streaming detection, cross-platform support, and content security policy compliance.
(4) A web-based dataset sharing portal, analogous to DataShop for tutoring systems, where researchers can publish and access anonymized student-AI interaction datasets with consent-based data governance, enabling collaborative research across institutions.

2 Related Work

2.1 AI Usage Patterns in Education

Research on student-AI interaction has revealed concerning patterns. Students frequently request direct solutions rather than explanations [7], and many lack the metacognitive awareness to use AI as an effective learning partner [17]. Sheese et al. [15] analyzed help-seeking patterns with CodeHelp, an LLM-powered programming assistant with guardrails that prevent direct code generation [9], finding that students' help-seeking strategies varied significantly and that some students attempted to circumvent guardrails to obtain direct answers. Similarly, Raihan et al. [14] showed that existing LLM guardrails in CS education can be bypassed, motivating the need for more robust intervention approaches. Furthermore, studies relying on self-reported data or custom interfaces may not capture authentic usage patterns: students significantly underreport AI usage due to social desirability bias [10], and tend to behave differently in controlled settings. These limitations highlight the need for unobtrusive data collection tools that can observe natural AI usage behavior.

2.2 Prompt Engineering and Scaffolding

Prior work has explored teaching students to write better prompts for AI tools. Denny et al. [3] introduced "Prompt Problems," a novel exercise type where students practice crafting effective prompts for LLM code generation, demonstrating the importance of prompt literacy in CS education. However, most approaches rely on explicit instruction sessions or dedicated workshops rather than in-context guidance during actual AI use. Real-time scaffolding during authentic AI interactions remains underexplored, despite evidence that metacognitive prompts delivered at the point of need are more effective than pre-training alone [2]. Ma et al. [11] studied student-AI interactions in programming education and found that scaffolding metacognition during AI use promotes active engagement over passive answer-seeking. Similarly, Zhang et al. [18] proposed constraining LLM responses to encourage student inquiry rather than direct answer provision. Pian et al. [12] demonstrated that LLM-enhanced tutoring systems can shift students from "receiving fish" (getting answers) to "learning to fish" (developing problem-solving skills). Our approach brings these principles into the student's existing AI workflow, providing guidance at the exact moment students are formulating queries to commercial chatbots. This builds on the well-established distinction in educational psychology between answer-seeking and help-seeking, with the latter recognized as a self-regulated learning strategy associated with better outcomes [5].

2.3 Learning Analytics and Privacy

Collecting authentic student-AI interaction data at scale presents a fundamental tension between research utility and privacy protection. Existing approaches each have significant limitations. Platform-level logs (e.g., from ChatGPT or Claude) are inaccessible to external researchers due to commercial restrictions. Building custom AI interfaces [9] provides full data access but sacrifices ecological validity, as students may behave differently outside familiar tools. Self-report methods such as surveys and interviews suffer from recall bias and social desirability effects [10], while screenshot-based

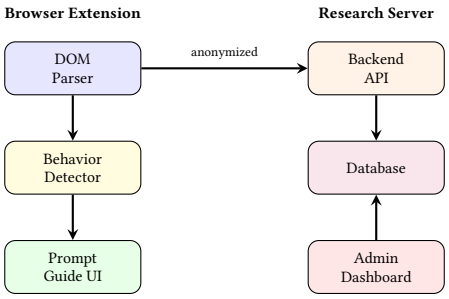

Figure 2: PromptTutor architecture. The extension captures conversations via DOM parsing, detects answer-seeking behavior locally, and sends anonymized data to the research server.

approaches are labor-intensive and difficult to scale. API-based collection requires researchers to host their own LLM infrastructure, introducing substantial costs and technical barriers.

Browser extensions offer a promising middle ground: they can observe authentic interactions on commercial platforms without requiring platform cooperation or custom infrastructure. Prior work has demonstrated this approach for tracking web browsing behavior in online courses [8]. PromptTutor extends this paradigm to AI conversation platforms, enabling researchers to collect naturalistic data from thousands of students at zero marginal cost per participant, a key requirement for learning at scale research.

However, collecting conversation data raises significant privacy concerns. Students may share sensitive personal information, academic struggles, or mental health concerns with AI chatbots, making informed consent and data protection critical. Ethical frameworks such as DELICATE [4] emphasize transparency, consent, and data minimization as core requirements for learning analytics systems. PromptTutor operationalizes these principles through multiple layers: client-side SHA-256 hashing ensures that researchers never see students' email addresses; per-conversation consent toggles give students granular control over what is shared; a data transparency dashboard lets students view and permanently delete any collected data; and all data transmission uses encrypted channels. This privacy-by-design approach prioritizes learner trust and algorithmic transparency, aligns with FERPA and GDPR requirements, and makes PromptTutor suitable for deployment across institutional contexts.

3 System Design

PromptTutor consists of a Chrome browser extension, a backend API server, and a web-based administration interface. The full source code is available at https://anonymous.4open.science/r/PromptTutor. Figure 2 illustrates the overall architecture and Figure 1 shows the user-facing interface.

3.1 Function 1: Privacy-Preserving Data Collection

3.1.1 Anonymous Identification. When users authenticate via Google OAuth, their email address is hashed using SHA-256 *on the client side* before any data transmission. The backend server never receives actual email addresses. This one-way hash serves as an anonymous identifier enabling longitudinal tracking without compromising identity.

3.1.2 Multi-Layer Consent. We implement a comprehensive consent model: (1) *Initial consent*: users agree to terms explaining data collection purposes, the possibility that anonymized data may be shared with other researchers for secondary analysis, and their rights. (2) *Real-time toggle*: a floating overlay on supported AI platform pages allows enabling or disabling sharing per conversation with a single click. (3) *Retroactive deletion*: turning off the toggle automatically deletes the current conversation's data from the server. (4) *Full transparency*: users can view and delete all their uploaded data at any time.

3.1.3 Data Capture. The extension uses DOM-based parsing with MutationObserver to capture conversations. This approach works within Content Security Policy constraints and handles LLM streaming output through debounced capture with 2-second stability detection. Captured data includes message content, timestamps, conversation grouping, and platform identifier. We exclude email addresses, authentication tokens, and browser metadata.

3.2 Function 2: Real-Time Prompt Guidance

The prompt guidance component operates entirely within the browser extension to minimize latency and preserve privacy.

3.2.1 Answer-Seeking Detection. When a student submits a message to a supported AI platform, the extension analyzes the message to classify it as *answer-seeking* or *help-seeking*. Answer-seeking queries request direct solutions, such as "Solve this equation for x: 3x + 5 = 20" or "Write a Python function that sorts a list." Help-seeking queries ask for explanations or conceptual understanding, such as "Can you explain how quicksort works?" or "I'm stuck on this step, what concept am I missing?"

The detection builds on prior work validating LLM-based behavioral labeling for student-AI dialogues (citation omitted for anonymous review), where a classification pipeline achieving $\kappa = 0.78$ agreement with human consensus was developed through iterative prompt refinement. By default, the extension uses a lightweight adaptation combining pattern matching on common answer-seeking indicators (imperative verbs like "solve," "write," "give me the answer") with rule-based classification optimized for real-time performance. We prioritize precision over recall: it is better to miss some answer-seeking queries than to incorrectly interrupt legitimate help-seeking behavior. For researchers seeking higher accuracy, PromptTutor also supports LLM-based classification by allowing researchers to configure their own API keys, enabling more nuanced detection at the cost of additional latency. Alternatively, researchers can export collected data as CSV for offline analysis using custom classifiers or coding schemes.

3.2.2 Scaffolding Intervention. When answer-seeking behavior is detected, the extension displays a non-blocking overlay suggesting the student reformulate their query. For example, if a student types "Write a function to reverse a linked list," the overlay might suggest: "Try asking: 'What are the key steps in reversing a linked list?' or

'Can you walk me through the logic?' " The student can accept the suggestion, dismiss it, or proceed with their original query. All interactions with the guidance system, including dismissals, are logged for research analysis when data collection is enabled.

Figure 1 illustrates the three main components of the PrompTutor interface. The extension popup (left) allows students to authenticate, toggle data sharing consent, and select between data collection only or data collection with prompt guidance. During a ChatGPT session (center), a floating indicator confirms active data collection. When the system detects an answer-seeking query (right), a scaffolding overlay appears with an explanation of why the query may limit learning, a suggested rephrasing that encourages deeper engagement, and three response options: use the suggestion, edit the original prompt, or send the query as-is.

3.2.3 Design Principles. The guidance system follows several key principles:

- **Non-blocking**: suggestions appear as overlays, never preventing the student from submitting their original query.
- **Constructive**: suggestions model better prompting strategies rather than simply telling students what not to do.
- **Adaptive**: if a student repeatedly dismisses guidance, the frequency of prompts is reduced to avoid frustration.
- **Transparent**: students can disable guidance independently of data collection.

4 Implementation

The Chrome extension uses Manifest V3 with vanilla JavaScript to minimize dependencies. The backend is Node.js with Express, using PostgreSQL for storage.

Several technical challenges required careful solutions:

Content Security Policy. ChatGPT enforces strict CSP that blocks external script injection. Our initial API interception approach would have provided cleaner structured data, but we pivoted to DOM parsing, which operates within the page's security constraints while still capturing essential conversation content.

Streaming Detection. AI responses stream incrementally, triggering hundreds of DOM mutations for a single message. Naive approaches would either miss content or create duplicate records. Our 2-second debounce with content length stability checking captures only complete responses while handling edge cases like user interruptions.

Cross-Platform Support. Each supported platform has different DOM structures and updates its interface frequently. We implemented a multi-selector fallback system that tries multiple CSS selectors in order, providing resilience against UI changes.

Low-Latency Detection. The behavior classifier must run fast enough to provide guidance before the AI begins generating a response. We use lightweight pattern matching that completes within milliseconds, avoiding the latency of API-based classification.

Extension Context Invalidation. When the extension updates or reloads, existing content scripts become invalid. We implemented graceful error handling with silent failures to prevent console errors from confusing users.

The administrator interface allows researchers to browse anonymized conversations grouped by user hash, filter by date range and platform, analyze prompt guidance interaction patterns (acceptance and dismissal rates, query reformulation patterns), and export data in JSON format for analysis with standard tools.

5 Planned Evaluation

We plan to evaluate PrompTutor through a controlled classroom study in an introductory programming course over one academic semester. Students will be randomly assigned to either a treatment group (data collection with prompt guidance) or a control group (data collection only), allowing us to measure the causal effect of real-time scaffolding on answer-seeking behavior and learning outcomes. Key measures include the proportion of answer-seeking vs. help-seeking queries, guidance acceptance and dismissal rates, and academic performance (assignments, exams). The study will follow the DELICATE ethical framework [4] with voluntary participation, granular consent, and full data transparency.

6 Conclusion and Future Work

PrompTutor addresses two interconnected challenges in AI-assisted education: the difficulty of collecting authentic student-AI interaction data, and the prevalence of ineffective AI usage patterns among students. By combining privacy-preserving data collection with real-time scaffolding in a single browser extension, our system serves as both a research infrastructure and an educational intervention.

A key advantage of PrompTutor's design is that it works with students' existing AI tools rather than requiring a separate platform. Unlike custom-built tutoring systems [9, 12] that require students to switch away from familiar interfaces, PrompTutor integrates directly into the platforms students already use, preserving ecological validity while enabling both observation and intervention. This approach is zero-cost for researchers: no API fees, no infrastructure for hosting an LLM, and no changes required from the AI platforms themselves.

PrompTutor also serves as a dataset creation platform. With student consent, researchers can share their collected datasets with the broader research community, enabling secondary analysis and cross-institutional studies without requiring each research group to independently recruit participants and deploy data collection infrastructure. Additionally, PrompTutor enables practical applications in AI-permitted assessments, where instructors can evaluate students based on their AI interaction process rather than final outputs alone.

In future work, we plan to extend PrompTutor to support even more AI platforms as they emerge, explore adaptive scaffolding strategies that adjust intervention intensity based on individual learning progress, and investigate the long-term effects of prompt guidance on students' metacognitive skills and AI literacy. We also aim to study whether the scaffolding effect persists after the intervention is removed, that is, whether students internalize better prompting habits over time.

PrompTutor provides a hosted platform where researchers can register, create a study, and immediately begin collecting data without deploying any infrastructure. Students install the extension from the Chrome Web Store and enter a pairing code to join. The source code is available at https://anonymous.4open.science/r/PrompTutor for self-hosted deployments.

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